

2 The infectious disease cholera is caused by a bacterium.

(a) Fig. 2.1 shows a transmission electron micrograph of this bacterium.

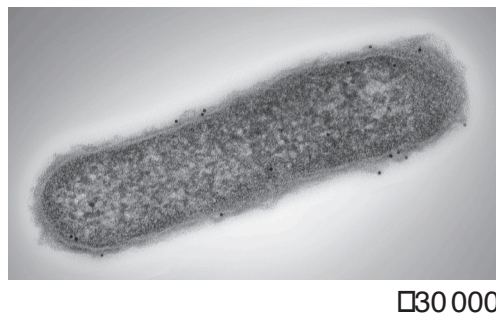


Fig. 2.1

(i) Name the bacterium that causes cholera.

..... [1]

(ii) The bacterium in Fig. 2.1 is an example of a prokaryotic cell.

Each of the descriptions **A** to **C** describes a cell structure found in prokaryotic cells **and** in plant cells.

For each of the descriptions **A** to **C**:

- name the cell structure described
- state **one** difference in this structure between a prokaryotic cell and a plant cell.

A the site of polypeptide synthesis

cell structure

difference

.....

B the genetic material of the cell

cell structure

difference

.....

C the structure that provides a rigid shape to the cell and prevents osmotic lysis

cell structure

difference

.....

[6]

(b) Cholera is an example of an infectious disease.

Explain what is meant by an infectious disease.

.....

.....

.....

..... [2]

The symptoms of cholera are caused by cholera toxin, a toxin released by the bacterium.

Cholera toxin is a protein made up of six polypeptides:

- a single copy of a polypeptide known as the A subunit that includes an extended alpha helix
- five polypeptides that together make the B subunit.

The B subunit of cholera toxin binds to a cell surface membrane component, known as GM1, of an intestinal epithelial cell. The complete cholera toxin protein then enters the cell by endocytosis. Once inside the cell, the A subunit of the protein acts as an enzyme, disrupting the normal functioning of the cell.

(c) List the levels of protein structure present in cholera toxin.

.....

..... [2]

(d) Outline the mechanism by which cholera toxin enters the cell.

You may use the space for annotated diagrams.

.....

.....

.....

.....

.....

..... [3]

(e) Using genetic engineering, it is possible to produce a form of cholera toxin consisting of only subunit B. This can be combined with inactivated bacterial cells to produce a vaccine against cholera.

(i) Suggest why subunit B, rather than subunit A, is used in the vaccine.

.....

.....

.....

..... [1]

(ii) Outline how this vaccine can give protection against cholera.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 20]